

GTSC2143 Machine Learning for Business

Tutorial 11

Please write down your answers in this document and submit it at iSpace by the end of this tutorial.

Activity 1. Data Loading and Processing

1. Load the Dataset

```
import pandas as pd
import numpy as np

np.random.seed(42) # set random seed

users = [f"U{str(i).zfill(3)}" for i in range(1, 21)]
items = [f"S{str(i).zfill(3)}" for i in range(1, 31)]

# generate data
num_rows = 1000
df = pd.DataFrame({
    "user_id": np.random.choice(users, num_rows),
    "item_id": np.random.choice(items, num_rows),
    "listen_count": np.random.randint(1, 6, num_rows)
})
```

2. Explore the dataset:

- a) Check the data shape
- b) Display the first 3 rows

Activity 2. Popularity-Based Recommendation

1. Compute the total listen count for each song (item)
2. Display the top 10 most frequently played songs

Activity 3. Item-Based Collaborative Filtering

1. Construct a song co-occurrence matrix
2. Compute an item similarity matrix using cosine similarity
3. Display the top 3 songs to song S003

- End of Tutorial 11 -